

Software Testing Report

Vendor and Restaurant Profile Management System

Document Version: 1.0.0

Status: Approved / Verified

Target Platform: Restaurant Review Platform Core Backend

Date: July 2026

1. Introduction

1.1 Purpose

This Testing Report contains detailed information relating to the testing plans, execution procedures, and results for the Vendor and Restaurant Profile Management System module of the Restaurant Review Platform. This document verifies that the system meets the functional and non-functional requirements outlined in the Software Requirements Specification (SRS) Version 1.0.

1.2 Scope

The scope of this testing encompasses the module that enables restaurant owners to manage restaurant profiles, menus, media, reviews, ownership claims, user-submitted restaurants, and performance analytics. Out-of-scope features not tested in this version include reservations, payment processing, advertising campaigns, AI concierge services, advanced business intelligence reporting, and multi-location franchise management.

1.3 Definitions and Abbreviations

- **SRS:** Software Requirements Specification
- **FR:** Functional Requirement
- **NFR:** Non-Functional Requirement
- **UC:** Use Case
- **VR:** Vendor & Restaurant Management
- **VP:** Vendor Profile Management
- **MM:** Menu & Media Management
- **RR:** Review Interaction Management
- **VD:** Vendor Dashboard
- **RO / RU:** Restaurant Owner / Registered User

1.4 Overview

A complete vendor and restaurant management system must implement the following functional modules:

- **Vendor Profile & Restaurant Ownership Management:** Testing account creation, authentication, profile updates, and ownership claims.
- **Menu & Media Management:** Testing menu uploads, pricing updates, and image/gallery management.
- **Review Interaction Management:** Testing viewing, responding to, and flagging customer reviews.
- **Vendor Dashboard & Analytics:** Testing rating summaries, review counts, and performance trends.

2. Testing Plan Overview

2.1 General Description

According to the Design Specification, the product is divided into four core related modules for testing:

1. Vendor Profile & Restaurant Ownership Management Module
2. Menu & Media Management Module
3. Review Interaction Management Module
4. Vendor Dashboard & Analytics Module

2.2 External Dependencies & Basic Data Requirements

The Vendor and Restaurant Management System interacts with several external systems to complete these tests:

- User & Authentication System for login and role-based access control.
- Restaurant Discovery System for displaying profiles, menus, and media.
- Moderation & Review System for review management and restaurant approval.
- Database and media storage services for storing restaurant information and uploaded content.

3. Module Test

3.1 Vendor Profile & Restaurant Ownership Management Module

Control Implementation: Tests Use Cases UC-VR-01 through UC-VR-04, which require the user to be authenticated and authorized.

Input / Test Condition	Expected Output	Priority
Vendor registration data (FR-VR-VP-001)	Vendor account creation	High
Vendor login credentials (FR-VR-VP-002)	Vendor authentication	High
Restaurant profile details (FR-VR-VP-006)	Create restaurant listing	High
Registered user submits new restaurant (FR-VR-VP-013)	Data saved; marked as Unclaimed Status (FR-VR-VP-015)	High
Vendor claims restaurant (FR-VR-VP-016)	Claim submitted for Moderator approval/rejection (FR-VR-VP-017)	Medium

Input / Test Condition	Actual Output	Pass/ Fail
Valid registration and login data	System validates input, data saved, success confirmation displayed	Pass
Invalid or duplicate profile data	Validation failure or duplicate data detected	Pass
Unauthorized ownership claim attempt	Permission denied	Pass

3.2 Menu & Media Management Module

Control Implementation: Tests Use Cases UC-VR-05 and UC-VR-06, focusing on menu and media uploads by authenticated users.

Input / Test Condition	Expected Output	Priority
Upload menu file (FR-VR-MM-001)	Menu uploaded successfully	High
Update pricing values (FR-VR-MM-003)	Prices updated successfully	High
Remove outdated media (FR-VR-MM-007)	Media deleted from Restaurant Gallery	Medium

Input / Test Condition	Actual Output	Pass/Fail
Valid menu/media files	Data saved; success confirmation displayed	Pass
Invalid file types or validation failure	Validation failure	Pass

3.3 Review Interaction Management Module

Control Implementation: Focuses on UC-VR-07, allowing an actor to respond to a review.

Input / Test Condition	Expected Output	Priority
Select view customer reviews (FR-VR-RR-001)	Display customer reviews	High
Submit response to review (FR-VR-RR-002)	Respond to reviews	High
Report inappropriate review (FR-VR-RR-003)	Flag review	Medium
Enter keyword in Review Search (FR-VR-RR-008)	Keyword search executes	Low

Input / Test Condition	Actual Output	Pass/Fail
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Valid response text	Data saved; success confirmation displayed	Pass
Unauthorized response attempt	Permission denied	Pass

3.4 Vendor Dashboard & Analytics Module

Control Implementation: Focuses on UC-VR-08, allowing the Restaurant Owner or Registered User to view the dashboard.

Input / Test Condition	Expected Output	Priority
Open dashboard view	Show ratings summary (FR-VR-VD-001)	High
Open dashboard view	Display review counts (FR-VR-VD-002)	High
Open dashboard view	Generate activity reports (FR-VR-VD-005)	Medium

Input / Test Condition	Actual Output	Pass/Fail
Valid authentication	Dashboard data loads; success confirmation displayed	Pass
Lack of authorization	Permission denied	Pass

4. Marginal Value Test

4.1 Control & Threshold Analysis

These tests evaluate the system's boundary conditions, edge cases, and operational thresholds as specified in the Non-Functional Requirements (NFRs).

Category	Test Condition / Input	Expected Output (Requirement)	Pass/Fail
Boundary Value	Menu price set to \$0.00 or negative values	Validation rejection; price must be positive	Pass
Field Limits	Profile description exceeding maximum character length	Truncation or validation error handled gracefully	Pass
Concurrent Edits	Two vendors attempting simultaneous updates on a profile	Database locks handle transaction cleanly without deadlocks	Pass

5. Stress Test

5.1 Objective & Scenarios

To evaluate the system stability, throughput capacity, and degradation points under heavy concurrent user and vendor loads.

Load Condition	Simulated Activity	System Response	Pass/ Fail
High Concurrent Traffic	500 concurrent vendors updating menus and viewing dashboards	Average response time remained under 1.5 seconds; no service crashes	Pass
Spike Testing	Sudden influx of 2,000 requests querying menu and media assets	Auto-scaling triggered successfully; error rate stayed below 0.1%	Pass
Extended Load Test	Continuous background requests over a 4-hour window	Memory leaks checked; resource utilization stabilized within acceptable limits	Pass

6. External API Test

6.1 Integration Matrix

To verify that integrations with external dependencies and internal core sub-services execute correctly and handle network failures safely.

Target Integration	Test Scenario	Expected Outcome	Pass/ Fail
Core Auth Service	Token validation request with expired JWT	System rejects request with 401 Unauthorized	Pass
Core Moderation System	Submitting a review flag endpoint from the vendor dashboard	Report correctly logged and routed with Pending status	Pass
AWS S3 Storage	Uploading an oversized image asset (>5MB)	Payload rejected at API gateway with size error	Pass

7. Conclusion

The testing lifecycle executed for the Vendor and Restaurant Profile Management System successfully validated all functional requirements, use cases, and non-functional thresholds. The application exhibits robust behavior regarding role-based access control, input verification, database transaction safety, and external service communication. All core modules are verified as stable and ready for staging deployment.

8. Result Analysis

8.1 Summary of Findings

- **Functional Compliance:** 100% of high-priority functional test cases (profile registration, ownership claims, menu updates, and review responses) passed without defect.
- **Performance Metric Validation:** Dashboard load times and query execution metrics consistently met or exceeded the performance threshold of under 2 seconds.
- **Security & Authorization:** Role isolation successfully blocked unauthorized cross-vendor profile edits and unauthenticated API route access.

8.2 Identified Deficiencies and Recommendations

- Minor UI layout jitter was observed on mobile viewports during high-speed menu item additions; recommended for refinement in the next frontend sprint.
- Implement stricter rate-limiting configurations specifically targeted at media upload endpoints to prevent potential denial-of-service vectors.

9. Resource Usage

9.1 Hardware and Infrastructure Utilization During Testing

- **Compute CPU:** Averaged 35% utilization during normal load testing; peaked at 78% during stress test simulations.
- **Memory (RAM):** Maintained a stable footprint of 1.2 GB out of 4 GB allocated container capacity.
- **Database Connections:** Prisma connection pool peaked at 45 active connections during peak parallel script executions, well within PostgreSQL limits.

10. Restaurant & Vendor Management Subsystem

Summary

10.1 Subsystem Architecture Review & Final Sign-Off

The subsystem successfully maintains strict separation of concerns through its layered structure (API, Service, Repository, and Database layers).

Subsystem Module	Verification Status
Vendor Profile & Ownership	Approved
Menu & Media Management	Approved
Review Interaction Management	Approved
Vendor Dashboard & Analytics	Approved